

1. Find your Public IP Address

- Search "What is my IP" on Google.

The screenshot shows the homepage of WhatIsMyIP.com. The navigation bar includes links for Tools, Videos, News, Pricing, API, Sign Up, Login, and Help. The main menu features links for What Is My IP?, IP Address Lookup, IP Whois Lookup, Speed Test, IP Tools, Privacy & Security, and Guides. The 'What Is My IP?' section displays the user's IP address as 152.59.19.174 and their IPv6 address as 2409:40c1:200f:e353:a51a:eca6:f716:4956. Below this, there are tabs for IPv4 Details and IPv6 Details. The IPv4 Details tab is active, showing a warning 'VISIBLE FROM YOUR CONNECTION' and a list of location details: City: Batwa, State/Region: Gujarat, Country: India, Postal Code: 380028, Time Zone: UTC +05:30, ISP: Reliance Jio Infocomm Limited, and ASN: 55836. A red button at the bottom of the details section says 'Protect My Privacy with a VPN'. To the right of the details is a map of Ahmedabad, India, with a link to 'Open in Maps'.

2. Find the IP Address of the following websites:

- google.com
- facebook.com
- amazon.com

Active IP addresses assigned to these domains include:

- [Google.com](https://google.com): 142.250.190.46
- [Facebook.com](https://facebook.com): 157.240.224.35
- [Amazon.com](https://amazon.com): 54.239.25.207

- Finding ip address through command prompt:

```
C:\Users\tusha>ping google.com

Pinging google.com [2404:6800:4002:802::200e] with 32 bytes of data:
Reply from 2404:6800:4002:802::200e: time=103ms
Reply from 2404:6800:4002:802::200e: time=108ms
Reply from 2404:6800:4002:802::200e: time=106ms
Reply from 2404:6800:4002:802::200e: time=102ms

Ping statistics for 2404:6800:4002:802::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 102ms, Maximum = 108ms, Average = 104ms
```

```
C:\Users\tusha>ping facebook.com

Pinging facebook.com [2a03:2880:f311:1:face:b00c:0:25de] with 32 bytes of data:
Reply from 2a03:2880:f311:1:face:b00c:0:25de: time=103ms
Request timed out.
Reply from 2a03:2880:f311:1:face:b00c:0:25de: time=105ms
Reply from 2a03:2880:f311:1:face:b00c:0:25de: time=99ms

Ping statistics for 2a03:2880:f311:1:face:b00c:0:25de:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 99ms, Maximum = 105ms, Average = 102ms
```

```
C:\Users\tusha>ping amazon.com

Pinging amazon.com [64:ff9b::6252:a1b9] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 64:ff9b::6252:a1b9:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Amazon and facebook requests are timed out because many large websites block or ignore ping requests for security reasons. Even though the website is working normally in a browser, it may not reply to ICMP ping packets.

3. Explain the difference between:

- HTTP and HTTPS
- Public IP and Private IP

Difference Between HTTP and HTTPS

HTTP	HTTPS
HTTP stands for HyperText Transfer Protocol. In HTTP, the URL begins with “http://”.	HTTPS stands for HyperText Transfer Protocol Secure. In HTTPS, the URL starts with “https://”.
HTTP uses port number 80 for communication.	HTTPS uses port number 443 for communication.
Hyper-text exchanged using HTTP goes as plain text i.e. anyone between the browser and server can read it relatively easily if one intercepts this exchange of data and due to which it is Insecure.	HTTPS is considered to be secure but at the cost of processing time because Web Server and Web Browser need to exchange encryption keys using Certificates before actual data can be transferred.
HTTP Works at the Application Layer .	HTTPS works at Transport Layer .
HTTP does not use encryption, which results in low security in comparison to HTTPS.	HTTPS uses Encryption which results in better security than HTTP.

HTTP speed is faster than HTTPS.	HTTPS speed is slower than HTTP.
HTTP does not use data hashtags to secure data.	HTTPS will have the data before sending it and returning it to its original state on the receiver side.
HTTP is used to transfer text, video, and images via web pages.	HTTPS is used to transfer data securely via a network.

Difference Between Public IP and Private IP

Private IP Address	Public IP Address
Used within a local or private network	Used for communication over the internet
Not routable on the public internet	Routable on the public internet
Scope is limited to the local network	Scope is global
Assigned by router or DHCP server	Assigned by ISP
Unique within a local network	Globally unique

Requires NAT for internet access	Does not require NAT
Hidden from external networks	Visible on the internet
Uses reserved private IP ranges	Uses globally assigned IP ranges
Freely usable inside networks	May involve additional ISP cost
Example: 192.168.1.10	Example: 203.0.113.10

4. Open any 5 websites and check:

- Do they use HTTPS?
- What company owns the website?

Website	Uses HTTPS?	Owner / Parent Company
GeeksforGeeks	Yes	Sanchhaya Education Pvt. Ltd. (Founded by Sandeep Jain)
ChatGPT	Yes	OpenAI
GitHub	Yes	Microsoft
Reddit	Yes	Reddit, Inc. (Public company)
Netflix	Yes	Netflix, Inc. (Public company)